

CSCE 671 HUMAN-COMPUTER INTERACTION · MODULE 03 · LECTURE NOTES

Attention, Memory, and Error

Hard limits, and the error rates they imply.

MODULE OUTCOMES

- Explain attention as a limited resource with a switching cost.
- Explain working memory's limit and its design consequences.
- Explain recognition against recall.
- Classify errors as slips or mistakes and fix each accordingly.
- Predict error rates from design properties.

1 Attention

ATTENTION IS A SINGLE CHANNEL, AND SWITCHING IT COSTS TIME AND ACCURACY

People do not multitask; they switch between tasks — and every switch costs reorientation time and raises the subsequent error rate, both of which are measured and substantial rather than marginal.

So an interruption is expensive in a way its duration badly understates: a five-second notification can cost far more than five seconds of task progress, because the mental state of the interrupted task has to be rebuilt from whatever is still visible.

And the cost scales with the interrupted task's complexity — which means interruptions are worst precisely when the user is doing something that matters and has the most state to lose.

Which gives a clear design rule: interrupt only for something more important than whatever the user is currently doing — and the system does not know what that is, so the default should be not to interrupt and to let the user come to the information.

- **Batch notifications** rather than delivering each one as it arrives — **which converts many switches into one** and is the single largest available improvement to a notification system.
- **Make interruptions resumable: preserve the state, the scroll position, the selection, and any partially entered data — because losing them makes the switching cost effectively unbounded** rather than merely high.
- **And do not steal focus. A dialogue that appears under the cursor while somebody is typing receives input that was intended for something else — which is a destructive failure rather than an annoyance**, since the keystroke may have dismissed it with a choice nobody made.
- **Reserve modal interruption for the genuinely irreversible**, which on inspection is almost nothing (§4, and Module 01 §4's undo preference).

- **And note that this is the same argument as CSCE 701 Module 11's compliance budget — attention is finite and everything spends from it**, so each interruption reduces the attention available for whatever you most wanted them to notice. **Focus stealing is the one that causes actual data loss**, and it is still common, which makes it worth naming as a defect rather than a preference.

2 Working memory

WORKING MEMORY HOLDS ABOUT FOUR CHUNKS, NOT SEVEN, AND LOSES THEM IN SECONDS

The famous figure of seven plus or minus two is higher than the current estimate — the modern figure for unrehearsed capacity is around four chunks, and it falls further under any concurrent task, including the task of operating your interface.

And a chunk is whatever the person has already learned as a single unit — so an expert holds more of their own domain because their chunks are larger, not because their memory capacity is greater, which is the finding that makes expertise intelligible.

Which means any design requiring a person to carry information between screens is requiring something they mostly cannot do — a confirmation code from one page to another, a value to compare against, their position in a multi-step sequence.

So the rule is: show it, do not ask them to remember it — which is almost always technically possible (the system has the information) and is the most frequently violated principle in this course.

- **Keep the information needed for a decision visible at the moment of the decision — not on a previous screen**, which is where it usually is because that is where it was collected.
- **Avoid modal dialogues that hide the thing they are asking about** — 'are you sure you want to delete this?' over the top of the thing being deleted is a **common and self-defeating pattern**.
- **Prefer recognition to recall (§3) — show the available options rather than requiring the person to produce the name**, which converts a memory task into a matching task.
- **And do not require re-entry of anything the system already holds** — which is both a memory demand and an error opportunity, and is sometimes defended as verification when it is actually transcription.
- **Which is why a multi-step wizard that forgets what you typed on step two is worse than a single long form**: the form at least keeps everything visible. **'Show it, do not ask them to remember it'** resolves most of this module's design questions, and it is cheap, because the system already has the information — the only cost is layout.

3 Recognition and recall

RECOGNISING SOMETHING IS FAR EASIER THAN RECALLING IT, WHICH IS WHY MENUS BEAT COMMAND LINES FOR NOVICES

Recognition requires only matching against what is present; recall requires generating the answer from nothing — and **the difference in success rate is large**, consistently, across every domain it has been measured in.

So a visible list of options is usable immediately, and a command requiring the exact name is not — which is why **discoverability and menus go together**, and why a powerful tool with no visible surface has a learning cliff rather than a curve.

But recall is faster once it has been learned, and an expert using a command line outperforms the same person using menus by a wide margin — so the asymmetry does not by itself settle the design question, which is the part usually left out.

Which argues for both: a discoverable path for learning and a fast path for expertise, with the fast path's existence signified within the slow one — a menu item displaying its own keyboard shortcut, or a command palette that shows the shortcut for whatever you just ran. **This is the pattern that serves both populations without compromising either.**

4 Error

	Slip	Mistake
What happened	The right intention, executed with the wrong action.	The wrong intention, executed perfectly correctly.
The cause	Attention, similarity between controls, and automaticity.	A wrong mental model of the system (Module 04).
Who makes them	Experts, more than novices — see the note.	Novices, more than experts.
The fix	Differentiate the controls, separate them spatially, and make the action reversible.	Fix the conceptual model and the signifiers that convey it (Module 04 §3).
Example	Clicked the menu item adjacent to the intended one.	Believed that 'archive' deleted the message.

Experts make more slips than novices because expertise is automaticity — the expert is not attending to the individual actions, which is what makes them fast and what makes the occasional wrong one go uncaught. **Which means a design that is safe for novices may be dangerous for the people who use it most, and that is the counter-intuitive finding of the module:** the destructive control adjacent to a common one is a hazard specifically for your heaviest users.

AND THE ERROR RATE IS PREDICTABLE FROM THE DESIGN, WHICH MAKES IT AN ENGINEERING QUANTITY

Adjacent destructive and benign controls produce slips at a rate you can measure — and separating them lowers it — which is a design change with a predicted effect rather than a matter of preference.

Similarly, and each with a predictable rate: a required value held across screens produces transcription errors; an unlabelled unit produces magnitude errors; a modal dialogue produces reflexive dismissal; two similar labels produce selection errors.

So error is not a property of the user but a joint property of the person and the design — which means it can be designed down, and the amount by which is measurable (Module 07's testing, and Module 08's experiment).

Which is Module 01 §4's reframing with a number attached — and is what turns a design argument into a prediction you can test, which is the move that makes this subject an engineering discipline rather than a critical one.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Johnson — Designing with the Mind in Mind, chapters 7 through 11	§1 through §3, with the corrected memory figures and the sources. Library copy.
Cowan — The magical number 4 in short-term memory (free)	§2's correction in the original — why the figure is four rather than seven, and what a chunk is.
Norman — The Design of Everyday Things, chapter 5	§4's slip and mistake taxonomy, with the sub-categories and the fixes.
Mark, Gudith & Klocke — The cost of interrupted work (free)	§1's switching cost, measured — and the finding that people compensate by working faster, at a cost in stress.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Time yourself on a task with and without an interruption, and measure the recovery.
- 2 Find a notification in your own software that could be batched.
- 3 Find a focus-stealing dialogue and reproduce the data loss it can cause.
- 4 Find a place in an interface that requires carrying information between screens.

- 5 **Fix it by showing the information** at the point of decision.
- 6 **Count the chunks** a task in your software requires somebody to hold.
- 7 **Find a modal dialogue that hides what it asks about.**
- 8 **Add shortcut signifiers** to a menu, and check whether anybody learns them.
- 9 **Classify ten errors you have made** as slips or mistakes.
- 10 **Find adjacent destructive and benign controls** and predict the slip rate before measuring it.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why do people switch rather than multitask, and what does a switch cost?
- 2 Why does an interruption cost more than its duration?
- 3 Give five design consequences for attention, and the one that destroys data.
- 4 State the working memory limit, and what a chunk is.
- 5 What is the most frequently violated principle in the course?
- 6 Give four consequences of the memory limit.
- 7 State the recognition/recall asymmetry and why it does not settle the design.
- 8 Contrast slips and mistakes on five dimensions.
- 9 Why do experts make more slips, and what follows?
- 10 Why is error an engineering quantity?